

EE4C10

Analog Circuit Design

Fundamentals

Week 3

Qinwen Fan

LTSpice Simulation Session

- We have 3 mini exercises.
- Download the exercises from bright space:
<https://brightspace.tudelft.nl/d2l/le/content/422545/Home>

Exercise 1

- Open Exc3.asc, you will see a source follower with a load resistor.
- Run DC simulation to check whether the amplifier is biased properly
- Run a transient simulation to plot the output and input voltage. Explain what you see. Plot the current in the resistor.
- Replace the resistor by a current source (30 μA). Plot the output and the input voltage. Explain the difference you see in this case compared to the previous case.

Note: when you use a current source, you may see a negative voltage at the source of M_1 , explain why this is the case. Is this realistic?

Exercise 2

- Open Exc4.asc, you will see a CG stage amplifier.
- Run a DC operating point simulation and check whether the transistor is biased in saturation
- Calculate the “transimpedance” of the amplifier ($\Delta V_{out}/\Delta I_{in}$)
- Run an AC simulation and plot the transimpedance and the current in R_2 .
- Add a 100 k Ω resistor in parallel with the current source to mimic its output impedance. Repeat the above calculation and simulation. Plot the output voltage and the current in R_2 .

Exercise 3

- Open Exc5.asc and you will see a cascode stage.
- Run a DC operating point simulation to see if all the transistors are biased in saturation
- Calculate the gain of the amplifier and run AC simulation to verify